

Fundamentals of theory of computation I.

Lecture 6

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Suffix languages of languages

Definition

Finite automata A and A' are called **equivalent**, if $L(A) = L(A')$ holds.

Definition

Given a language $L \subseteq \Sigma^*$. $L_p := \{v \in \Sigma^* \mid pv \in L\}$ is called the **suffix language** of L **with respect to the word** $p \in \Sigma^*$.

I.e., L_p is the set of possible right-extension for p to get a word of the language L .

E.g., if $L = \{ab, aac, bc, aabc\}$, then $L_{aa} = \{c, bc\}$.

Properties:

- ▶ $(L_p)_q = L_{pq}$,
- ▶ $L_\varepsilon = L$,
- ▶ $\varepsilon \in L_p \iff p \in L$.

Suffix languages of automata

Definition

Let $A = (Q, \Sigma, \delta, q_0, F)$ be a deterministic finite automaton.

$L(A, q) := \{v \in \Sigma^* \mid qv \Rightarrow_A^* p, p \in F\}$ is called the **suffix language** of A **with respect to the state** $q \in Q$.

Alternative definition: $L(A, q) := \{v \in \Sigma^* \mid \delta(q, v) \in F\}$.

I.e., $L(A, q)$ would be the language accepted by A if q were the initial state of the automaton.

Properties:

- ▶ $L(A, q)_w = L(A, \delta(q, w)) \quad (w \in \Sigma^*),$
- ▶ $L(A, q_0) = L(A),$
- ▶ $\varepsilon \in L(A, q) \iff q \in F.$

Myhill–Nerode theorem

Myhill–Nerode theorem

Let $L \subseteq \Sigma^*$. Then $L \in \mathcal{L}_3 \iff |\{L_p\}_{p \in \Sigma^*}| < \infty$.

Proof:

$(\Leftarrow)$:

Assume, that $\{L_p\}_{p \in \Sigma^*}$ is a finite set. We construct a deterministic finite automaton with the set of states $\{L_p\}_{p \in \Sigma^*}$. It is a finite set by the assumption.

State L_p is expected to be the state after fully processing p , starting from the initial state.

$$A_L^{MN} := (\{L_p\}_{p \in \Sigma^*}, \Sigma, \delta, L_\varepsilon, F),$$

where $F := \{L_p \mid \varepsilon \in L_p\}$, and $\delta(L_p, t) := L_{pt}$.

Myhill–Nerode theorem

Properties of A_L^{MN} :

- 1) δ is well defined, i.e., $L_p = L_q$ implies $\delta(L_p, t) = \delta(L_q, t)$ for all $t \in \Sigma$.
 - $L_p = L_q \implies (L_p)_t = (L_q)_t \implies L_{pt} = L_{qt} \implies \delta(L_p, t) = \delta(L_q, t)$ proving the statement.
- 2) $\delta(L_p, u) = L_{pu}$ holds for all $u \in \Sigma^*$.
 - Follows from the definition of the extended transition function.
Can be checked by induction on the length of u .
- 3) $L_p \in F \iff p \in L$.

We have

$$u \in L(A_L^{MN}) \iff \delta(L_\varepsilon, u) \in F \iff L_{\varepsilon u} = L_u \in F \iff u \in L,$$

implying $L(A_L^{MN}) = L$.

Myhill–Nerode theorem

($\Rightarrow$) :

Let $L \in \mathcal{L}_3$. According to a previous theorem there exists a DFA $A = (Q, \Sigma, \delta, q_0, F)$ with $L(A) = L$.

Then $|\{L(A, q)\}_{q \in Q}| \leq |Q| < \infty$ holds, since Q is a finite set.

Let $u \in \Sigma^*$ be arbitrary and consider suffix language L_u . Then we have $L_u = (L(A, q_0))_u = L(A, \delta(q_0, u))$.

This implies

$$\{L_u\}_{u \in \Sigma^*} \subseteq \{L(A, q)\}_{q \in Q},$$

proving the theorem.

Myhill–Nerode theorem

Example 1

$$L = \{a^n b^n \mid n \in \mathbb{N}\}.$$

$$\text{Then } L_{a^k} = \{a^{n-k} b^n \mid n \geq k\}. \quad (k \in \mathbb{N})$$

$$b^i \in L_{a^k} \iff i = k,$$

So, we have infinite pairwise disjoint suffix languages implying $L \notin \mathcal{L}_3$.

Myhill–Nerode theorem

Example 2:

$L = \{w \in \{a, b\}^* : N_a(w) \text{ és } N_b(w) \text{ páros}\}.$

$L_u = ?$ Easy to see, that L_u equals to one of

$L_\varepsilon = \{w \in \{a, b\}^* : N_a(w) \equiv N_b(w) \equiv 0 \pmod{2}\},$

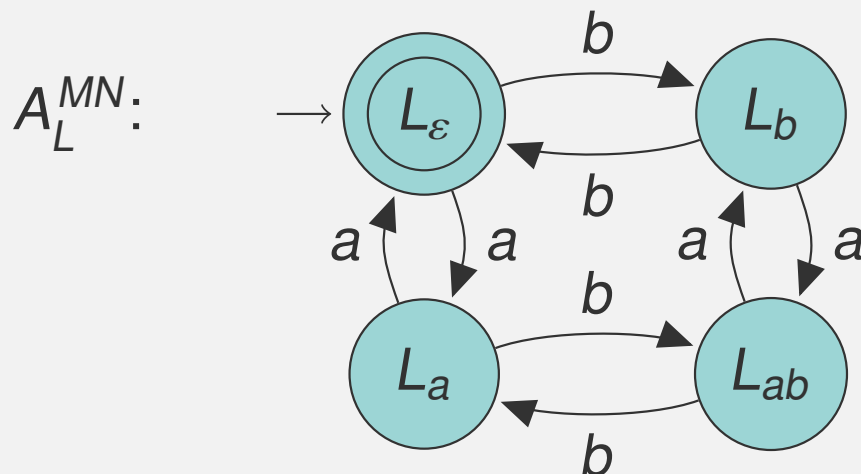
$L_b = \{w \in \{a, b\}^* : N_a(w) \equiv 0 \pmod{2}, N_b(w) \equiv 1 \pmod{2}\},$

$L_a = \{w \in \{a, b\}^* : N_a(w) \equiv 1 \pmod{2}, N_b(w) \equiv 0 \pmod{2}\},$

$L_{ab} = \{w \in \{a, b\}^* : N_a(w) \equiv N_b(w) \equiv 1 \pmod{2}\},$

according to the parity of $N_a(u)$ and $N_b(u)$.

E.g., $L_{abbabbbab} = L_a$. So we have $L \in \mathcal{L}_3$.



Minimal automata

Corollary

Let $L \in \mathcal{L}_3$ and let $A = (Q, \Sigma, \delta, q_0, F)$ be a DFA with $L(A) = L$. Then cardinality of the set of states for A_L^{MN} is less than or equal to $|Q|$.

Proof: The set of states of A_L^{MN} is $\{L_u\}_{u \in \Sigma^*}$ and we have $\{L_u\}_{u \in \Sigma^*} \subseteq \{L(A, q)\}_{q \in Q} \leq |Q|$ (last sentence of the proof of MN theorem).

Definition

A deterministic finite automaton A is called a (**minimal automaton**), if there is no other deterministic finite automaton A' with less states and $L(A') = L(A)$.

A minimal automaton A for a language $L \in \mathcal{L}_3$ is a minimal automaton with $L(A) = L$.

So A_L^{MN} is a minimal automaton for L .

Constructing minimal automata

Assume, that we have a DFA A . We give an algorithm constructing an equivalent minimal automaton. It has 2 steps.

1) Eliminating unreachable states

2) Contracting equivalent states

Two states q and q' are equivalent if the same set of words are accepted starting A from these states. For an input it does not matter whether we reach q or q' during computation, so these states can be contracted.

Observation: Two states q and q' can be contracted if and only if $L(A, q) = L(A, q')$ holds.

Eliminating unreachable states (reminder)

Definition

Let $A = (Q, \Sigma, \delta, q_0, F)$ be a deterministic finite automaton. A state $q \in Q$ is called **reachable** (from the initial state) if there is a reduction $q_0 x \Rightarrow^* q$ for some $x \in \Sigma^*$. Otherwise it is called **unreachable**.

Remark: q is reachable if and only if $\delta(q_0, x) = q$ holds for some $x \in \Sigma^*$.

Define a state H recursively as follows Let

$$H_0 = \{q_0\},$$

$$H_{i+1} = H_i \cup \{r \mid \delta(q, a) = r, q \in H_i, a \in \Sigma\}, \quad i = 1, 2, \dots$$

Since $H_0 \subseteq H_1 \subseteq \dots \subseteq H_k \subseteq Q$ holds, there must a natural number k with $H_k = H_{k+1}$. Then $H_\ell = H_k$ holds for every $\ell \geq k$. Let $H := H_k$. Then H is the set of reachable states.

Eliminating unreachable states (reminder)

Let us define a DFA $A' = (Q', \Sigma, \delta', q_0, F')$ as follows.

Let $Q' = H, F' = F \cap H, \delta' : H \times \Sigma \rightarrow H$ with the property $\delta'(q, a) = \delta(q, a)$, for every $q \in H$.

Example:

	a	b
$\rightarrow q_0$	q_2	q_4
$\leftarrow q_1$	q_3	q_0
$\leftarrow q_2$	q_0	q_2
q_3	q_1	q_2
q_4	q_5	q_2
$\leftarrow q_5$	q_4	q_2

$$H_0 = \{q_0\}$$

$$H_1 = \{q_0, q_2, q_4\}$$

$$H_2 = \{q_0, q_2, q_4, q_5\} =$$

$$H_3 = H$$

	a	b
$\rightarrow q_0$	q_2	q_4
$\leftarrow q_2$	q_0	q_2
q_4	q_5	q_2
$\leftarrow q_5$	q_4	q_2

Reduced automaton

Definition

Let $q, q' \in Q$ be states. q and q' are **indistinguishable**, if $L(A, q) = L(A, q')$ holds.

Notation: $q \sim q'$.

Observation:

- ▶ $\sim$ is an equivalence relation
- ▶ if $q \sim q'$ holds, then we have $\delta(q, t) \sim \delta(q', t)$ for all $t \in \Sigma$. ($\sim$ has the **right congruent** property)

Definition

We call a deterministic finite automaton **reduced**, if it has no indistinguishable states.

Factor automaton

Definition

Let $\sim$ be the indistinguishable relation of DFA $A = (Q, \Sigma, \delta, q_0, F)$. $A/\sim = (Q', \Sigma, \delta', q'_0, F')$, the **factor automaton** of A is defined as follows.

- ▶ Let Q' be the set of equivalence classes of Q by $\sim$.
 - ▶ If $t \in \Sigma$ and $r \in q'$ is a representative of the class $q' \in Q'$ then let $\delta'(q', t)$ be the class of $\delta(r, t)$.
 - ▶ Let q'_0 be the class of q_0 .
 - ▶ $F' = \{q' \in Q' \mid r \in F \text{ holds for some representative } r \in Q \text{ of } q'\}$.
-
- ▶ For the definition of δ' we can choose any representative of the class due to $\sim$ being right congruent.
 - ▶ If any representative of a class is in F , then all representatives of the class is in F as well (they are indistinguishable for ε).

Factor automaton

Definition

Let $A = (Q, \Sigma, \delta, q_0, F)$ and $A' = (Q', \Sigma, \delta', q'_0, F')$ be deterministic finite automata. A and A' are called **isomorphic**, if there is a bijection $\varphi : Q \rightarrow Q'$ with the properties

- ▶ $\varphi(q_0) = q'_0$.
- ▶ $\varphi(F) := \bigcup_{q \in F} \{\varphi(q)\} = F'$.
- ▶ $\forall q \in Q, t \in \Sigma$ we have $\varphi(\delta(q, t)) = \delta'(\varphi(q), t)$.

Theorem

The factor automaton $A/\sim$ of DFA A is

1. equivalent to A ,
2. reduced, and
3. if A has no unreachable states it is the only reduced DFA being equivalent to and isomorphic to A .

Factor automaton

Lemma 1

Let $q \in Q$ be an arbitrary state of A and let $q' \subseteq Q$ be the equivalence class of q by $\sim$. Then $L(A, q) = L(A/\sim, q')$ holds.

Proof: Let $u \in \Sigma^*$ be an arbitrary word. Then the equivalence class of $\delta(q, u)$ is $\delta'(q', u)$.

This can be proved by induction on $|u|$. The statement obviously holds in the case $|u| = 0$. Assume, that the statement holds for words shorter than $|u|$. Let $u = vt$ ($v \in \Sigma^*$, $t \in \Sigma$). We have $\delta'(q', vt) = \delta'(\delta'(q', v), t)$. Since v is shorter than u we know by induction that the equivalence class of $\delta(q, v)$ is $\delta'(q', v)$. Choose $\delta(q, v)$ as a representative of $\delta'(q', v)$. Then $\delta'(\delta'(q', v), t)$ is the equivalence class of $\delta(\delta(q, v), t) = \delta(q, vt) = \delta(q, u)$.

Then we have

$$u \in L(A, q) \Leftrightarrow \delta(q, u) \in F \Leftrightarrow \delta'(q', u) \in F' \Leftrightarrow u \in L(A/\sim, q').$$

Factor automaton

1. Apply Lemma 1 for q_0 .
2. Let $q', r' \in Q'$ be a pair of indistinguishable states of $A/\sim$, i.e., we have $q' \sim r'$. Then $L(A/\sim, q') = L(A/\sim, r')$ holds. According to Lemma 1 we have $L(A, q) = L(A, r)$ ($q \sim r$) for arbitrary representatives $q \in q', r \in r'$. It is possible only in the case $q' = r'$, so $A/\sim$ is a reduced automaton.
3. Follows from the next, more general lemma.

Lemma 2

Let $A = (Q, \Sigma, \delta, q_0, F)$ and $A' = (Q', \Sigma, \delta', q'_0, F')$ be reduced DFA's without unreachable states and being equivalent to each other. Then A and A' are isomorphic to each other.

The isomorphism:

$$\varphi(\delta(q_0, u)) := \delta'(q'_0, u)$$

for all $u \in \Sigma^*$. Since all states are reachable we have $\text{Dom}(\varphi) = Q$ and $\text{Ran}(\varphi) = Q'$.

Factor automaton

Let $u, v \in \Sigma^*$ be arbitrary words.

$$\begin{aligned} \delta(q_0, u) = \delta(q_0, v) &\stackrel{A \text{ is reduced}}{\iff} L(A, \delta(q_0, u)) = L(A, \delta(q_0, v)) \iff \\ L(A, q_0)_u = L(A, q_0)_v &\stackrel{\text{equivalence}}{\iff} L(A', q'_0)_u = L(A', q'_0)_v \iff \\ L(A', \delta'(q'_0, u)) = L(A', \delta'(q'_0, v)) &\stackrel{A' \text{ is reduced}}{\iff} \delta'(q'_0, u) = \delta'(q'_0, v). \end{aligned}$$

$(\Rightarrow)$ of first $\iff$ obviously holds, $(\Leftarrow)$ follows from A being reduced. Similarly, but for A' , in that case of the last $\iff$.

So φ is well defined and bijective.

- ▶ $\varphi(q_0) = \varphi(\delta(q_0, \varepsilon)) = \delta'(q'_0, \varepsilon) = q'_0,$
- ▶ $\varphi(F) = \bigcup_{u \in L(A)} \{\varphi(\delta(q_0, u))\} = \bigcup_{u \in L(A')} \{\delta'(q'_0, u)\} = F'$
- ▶ let $\delta(q_0, u) \in Q$ and $t \in \Sigma$ be arbitrary, then we have
 $\varphi(\delta(\delta(q_0, u), t)) = \varphi(\delta(q_0, ut)) = \delta'(q'_0, ut),$ and
 $\delta'(\varphi(\delta(q_0, u)), t) = \delta'(\delta'(q'_0, u), t) = \delta'(q'_0, ut),$

so φ is in fact an isomorphism proving Lemma 2 and the theorem.

An algorithm for state contraction

Notation: $\Sigma^{\leq i} := \{ w \in \Sigma^* \mid |w| \leq i \}$

Indistinguishability of q and q' means

$$\begin{aligned} q \sim q' &\iff L(A, q) = L(A, q') \iff \\ &\iff \forall u \in \Sigma^* : (\delta(q, u) \in F \iff \delta(q', u) \in F). \end{aligned}$$

Definition

$q \in Q$ and $q' \in Q$ are said to be **i -indistinguishable** ($i \geq 0$) (notation: $q \stackrel{i}{\sim} q'$), if

$$\forall u \in \Sigma^{\leq i} : (\delta(q, u) \in F \iff \delta(q', u) \in F).$$

Remark: i -indistinguishable states are also called i -equivalent to each other.

An algorithm for state contraction

Relation $\varrho_2 \subseteq X \times X$ is called a **refinement** of relation $\varrho_1 \subseteq X \times X$ if $x\varrho_2y \Rightarrow x\varrho_1y$ holds for every $x, y \in X$. (notation: $\varrho_1 < \varrho_2$),

Properties of $\stackrel{i}{\sim}$:

- ▶ $\stackrel{i}{\sim}$ is an equivalence relation,
- ▶ $q \stackrel{0}{\sim} q'$ holds if and only if $(q \in F \Leftrightarrow q' \in F)$,
- ▶ $q \stackrel{i+1}{\sim} q' \Leftrightarrow q \stackrel{i}{\sim} q' \wedge (\forall t \in \Sigma : \delta(q, t) \stackrel{i}{\sim} \delta(q', t))$ holds for all $q, q' \in Q$,
- ▶ $\stackrel{0}{\sim} < \stackrel{1}{\sim} < \stackrel{2}{\sim} < \dots < \sim$,
- ▶ $q \sim q' \Leftrightarrow \forall i \geq 0 : q \stackrel{i}{\sim} q'$.

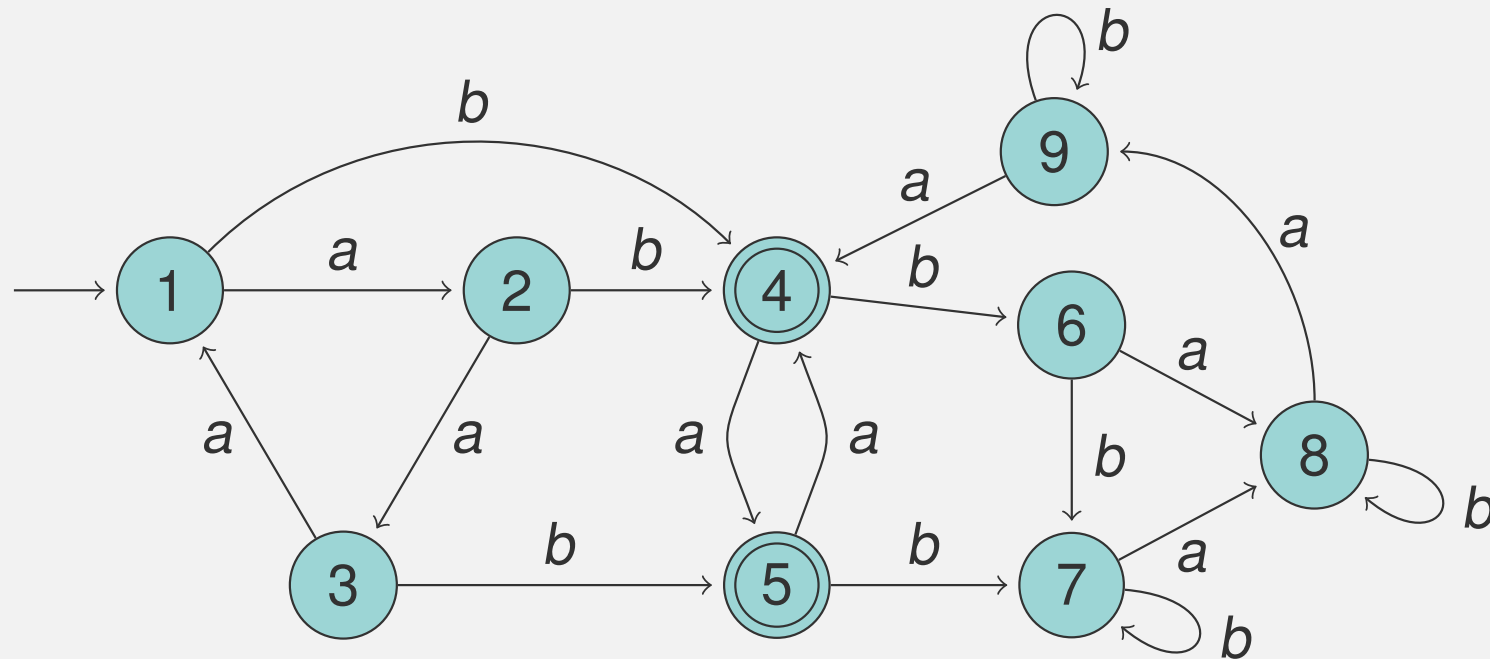
$\stackrel{i}{\sim}$ can be refined in at most $|Q| - 1$ steps.

$$i_0 := \min \{ i \mid \stackrel{i}{\sim} = \stackrel{i+1}{\sim} \}.$$

Then we have $i_0 \leq |Q| - 1$, and $\sim = \stackrel{i_0}{\sim}$.

An algorithm for state contraction

Example: Give a reduced automaton equivalent with the following automaton. (All states are reachable.)



An algorithm for state contraction

$\sim^0$: {1, 2, 3, 6, 7, 8, 9} , {4, 5}

	<i>a</i>	<i>b</i>
4	{4,5}	{1,2,3,6,7,8,9}
5	{4,5}	{1,2,3,6,7,8,9}

	<i>a</i>	<i>b</i>
1	{1,2,3,6,7,8,9}	{4,5}
2	{1,2,3,6,7,8,9}	{4,5}
3	{1,2,3,6,7,8,9}	{4,5}
6	{1,2,3,6,7,8,9}	{1,2,3,6,7,8,9}
7	{1,2,3,6,7,8,9}	{1,2,3,6,7,8,9}
8	{1,2,3,6,7,8,9}	{1,2,3,6,7,8,9}
9	{4,5}	{1,2,3,6,7,8,9}

$\sim^1$: {1, 2, 3} , {4, 5} , {6, 7, 8} , {9}

	<i>a</i>	<i>b</i>		<i>a</i>	<i>b</i>		<i>a</i>	<i>b</i>
4	{4,5}	{6,7,8}	1	{1,2,3}	{4,5}	6	{6,7,8}	{6,7,8}
5	{4,5}	{6,7,8}	2	{1,2,3}	{4,5}	7	{6,7,8}	{6,7,8}
			3	{1,2,3}	{4,5}	8	{9}	{6,7,8}

$\sim^2$: {1, 2, 3} , {4, 5} , {6, 7} , {8} , {9}

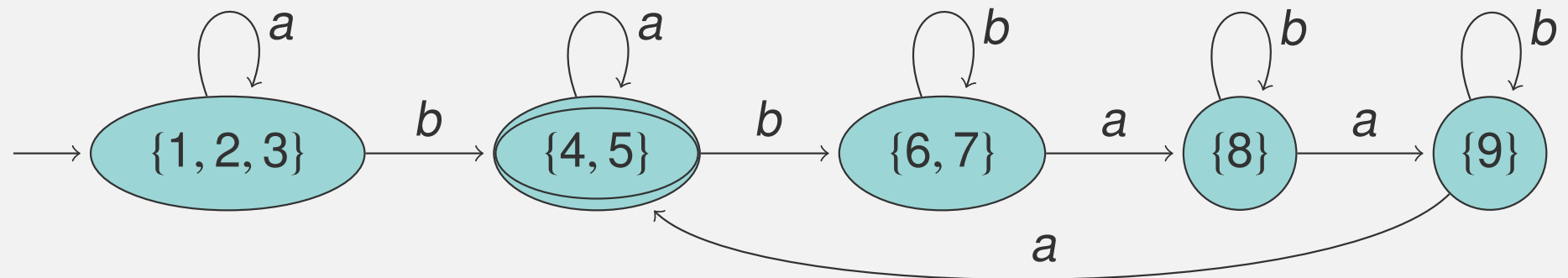
An algorithm for state contraction

$\tilde{2}$: $\{1, 2, 3\}$, $\{4, 5\}$, $\{6, 7\}$, $\{8\}$, $\{9\}$

	<i>a</i>	<i>b</i>		<i>a</i>	<i>b</i>		<i>a</i>	<i>b</i>
			1	$\{1, 2, 3\}$	$\{4, 5\}$			
4	$\{4, 5\}$	$\{6, 7\}$	2	$\{1, 2, 3\}$	$\{4, 5\}$	6	$\{8\}$	$\{6, 7\}$
5	$\{4, 5\}$	$\{6, 7\}$	3	$\{1, 2, 3\}$	$\{4, 5\}$	7	$\{8\}$	$\{6, 7\}$

$\tilde{3} = \tilde{2} = \sim$

The reduced automaton:



Any representative can be considered for determining transitions and accept states. New initial state is the class containing the original initial state.

Minimal automata – summary

We can construct a minimal automaton for DFA A in an algorithmic way by

1. eliminating unreachable states of A ,
 2. then contracting states by computing $\sim^i$ relations till they do not refine any more.
- ▶ This automaton has no unreachable states, reduced and equivalent with A and isomorphic to $A/\sim$,
 - ▶ Since A_L^{MN} , where $L = L(A)$ has no unreachable states, reduced (every minimal automaton is reduced) and equivalent with A we have that A_L^{MN} is isomorphic to the factor automaton,
 - ▶ Since A_L^{MN} is a minimal automaton, both the factor automaton and the automaton defined by the refinement of $\sim^i$'s are minimal automaton. Furthermore, according to our theorem they are isomorphic to each other.